



Software Testing Final Project ——AutoTestDesign

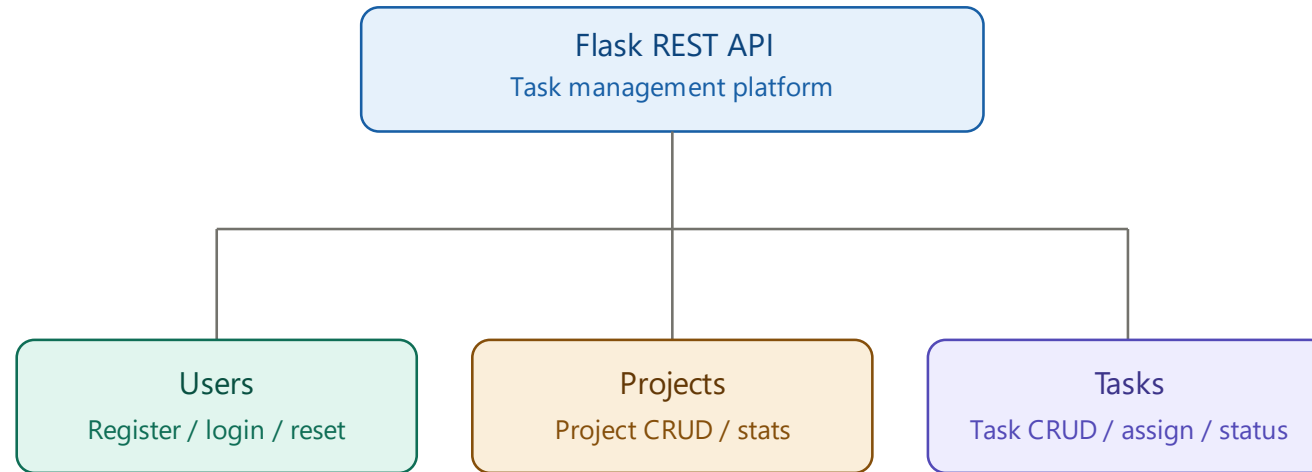
Group6

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Course Requirements & Implementation

ID	Function Category	Implementation
FR1/1.1	Requirement input & structuring	Extract fields, conditions, and expected behaviors from CSV / text / direct input
FR 2.0	Risk analysis & priority ranking	Generate a risk score, risk level, and test priority for each requirement
FR 3.0	Black-box test design	EP, BVA, and decision tables — three techniques to generate test cases
FR 4.0/5.0	White-box & oracle	Path coverage, state transition, expected-result generation
FR 6.0	Output & export	Generate in a structured format; export Pytest / JSON
FR 7.0	Test suite optimization	Risk-first prioritization, deduplication; designers can review and revise coverage items, strategies, and test cases

Test Target—— Task Management Platform



Why it is suitable for validating the tool

- It has both field validation and CRUD, assignment, statistics, and status behaviors; in addition, this project manually ported in some bugs, so it can measure whether the tool truly detects requirement deviations.

Workflow

1. Requirement Input

2. Risk Analysis

3. Black-box Test Design

4. White-box Test Design

5. Oracle Generation

6. Suite Optimization

7. Export

需求输入

代码仓库 · 需求文档 · 标准需求 & 需求 · 代码映射

代码仓库

路径

flask_app

13 个 .py 文件

需求文档

导入方式

☒ CSV 标准需求 ☐ TXT 文本需求 ☐ 粘贴文本

上传预定义的标准需求 CSV

task_pl_m_reqs.csv 3.1KB

```
id,title,description,input_fields,conditions,expected_behaviors,domains
REQ-001,用户注册,,["name","username","data_type","string",
"valid_range":"","3-20","constraints":["required","alphanumeric",
"unique"]],["name","password","data_type","string","valid_range":
"8-32","constraints":["required"]],["name","confirm_password",
"data_type","string","constraints":["required"]],["name","email"]]
```

解析到 7 条需求: REQ-001, REQ-002, REQ-003, REQ-004, REQ-005, REQ-006, REQ-007

生成标准需求与需求-代码映射

标准需求 (7)

REQ-001 用户注册 · 5 代码段

REQ-002 用户登录 · 1 代码段

输入字段

• username (string): --

required

• password (string): --

required

关联代码

routes/auth.py L126-186 | login()

条件 & 行为

• 账户已锁定

• [正确] → 登录成功

REQ-003 密码重置 · 2 代码段

REQ-004 项目管理 · 5 代码段

REQ-005 任务管理 · 5 代码段

REQ-006 任务分配 · 1 代码段

REQ-007 任务统计 · 1 代码段

Input:

Code directory / CSV requirement file /
TXT text / text

Output:

Standardized requirements, fields,
conditions, expected behaviors, requirement-
code mapping

Workflow

- 1. Requirement Input
- 2. Risk Analysis**
- 3. Black-box Test Design
- 4. White-box Test Design
- 5. Oracle Generation
- 6. Suite Optimization
- 7. Export

风险分析

LLM 评估每项需求的风险等级 & 测试优先级

开始风险分析

风险等级

High	Medium	Low	平均分
3	3	1	75.0

测试优先级

High	Medium	Low
3	3	1

需求	ψ_p 风险分数	ψ_r 风险	ψ_p 优先级	ψ_r 风险因素	ψ_r 原因
REQ-001	88	High	High	涉及密码验证; 敏感逻辑复杂; 安全关键功能	用户注册是系统的入口功能, 涉及密码和邮箱等敏感数据的验证, 任何缺陷都可能
REQ-002	92	High	High	涉及身份认证; 密码验证逻辑; 安全关键功能	用户登录是系统的核心功能, 涉及身份认证和安全性, 任何缺陷都可能导致用户无
REQ-003	85	High	High	涉及敏感数据; 密码重置流程复杂; 安全关键功能	密码重置功能涉及敏感数据和安全性, 任何缺陷都可能导致用户账户被盗或无法
REQ-004	70	Medium	Medium	涉及复杂业务逻辑; 多用户协作; 数据一致性要求高	项目管理功能是系统的重要业务模块, 虽然不涉及安全性, 但其复杂性和数据一致
REQ-005	65	Medium	Medium	涉及 CRUD 操作; 字段验证逻辑; 数据完整性	任务管理功能是系统的常用模块, 虽然不涉及安全性, 但其字段验证和数据完整性
REQ-006	75	Medium	Medium	涉及多用户操作; 权限控制逻辑; 数据一致性	任务分配功能涉及多用户协作和权限控制, 虽然不涉及安全性, 但其复杂性较高, 且
REQ-007	50	Low	Low	数据展示逻辑; 统计计算	任务统计功能主要是数据展示和统计计算, 功能失效影响较小或有替代方案, 可在

保存修改

Users can review, revise, and change risk levels and analyses

Workflow

- 1. Requirement Input
- 2. Risk Analysis
- 3. Black-box Test Design**
- 4. White-box Test Design
- 5. Oracle Generation
- 6. Suite Optimization
- 7. Export

黑盒测试设计

EP: BVA DT — LLM 生成分析表 & 测试用例

标准需求

REQ-005: 任务管理

测试技术

EP × BVA × DT ×

生成测试用例

共 18 用例

等价类划分 边界值分析 判定表 全部用例

等价类划分表

title		
类型	描述	代表值
valid	标题为非空字符串, 长度不超过200	有效标题
invalid	标题为空字符串	
invalid	标题长度超过200字符	a
invalid	标题为null	None

description

测试状态为null, 其他字段为有效值

```
{
  "title": "有效标题"
  "description": "任意描述"
  "priority": "Medium"
  "status": null
}
```

无效的状态值

测试用例审查与修改

ID	标题	描述	技术	类型	输入JSON
TC-EP-REQ-005-001	有效标题和默认优先级	测试标题为有效值, 优先级为默认值Medium, 状态为默认值todo	Equivalence Partitioning	Valid	{"title": "有效标题", "description": ""}
TC-EP-REQ-005-002	标题为空	测试标题为空字符串, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "", "description": "任意描述"}
TC-EP-REQ-005-003	标题超过最大长度	测试标题长度超过200字符, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "a", "description": ""}
TC-EP-REQ-005-004	标题为null	测试标题为null, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": null, "description": "任意描述"}
TC-EP-REQ-005-005	优先级为非默认值	测试优先级为非默认值, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "有效标题", "description": ""}
TC-EP-REQ-005-006	优先级为null	测试优先级为null, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "有效标题", "description": ""}
TC-EP-REQ-005-007	状态为非默认值	测试状态为非默认值, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "有效标题", "description": ""}
TC-EP-REQ-005-008	状态为null	测试状态为null, 其他字段为有效值	Equivalence Partitioning	Invalid	{"title": "有效标题", "description": ""}
TC-BVA-REQ-005-001	Title is empty	Test with title length less than minimum (0 characters)	Boundary Value Analysis	Boundary	{"title": "", "priority": "Medium", "status": "todo"}
TC-BVA-REQ-005-002	Title at minimum length	Test with title length at minimum (1 character)	Boundary Value Analysis	Boundary	{"title": "a", "priority": "Medium", "status": "todo"}

Select test requirements

Select test technique:EP BVA DT

Based on requirement fields, input constraints, and condition combinations, the system generates analysis tables and corresponding test cases

Manual review and modification

Workflow

1. Requirement Input
2. Risk Analysis
3. Black-box Test Design
- 4. White-box Test Design**
5. Oracle Generation
6. Suite Optimization
7. Export

白盒测试设计

路径覆盖 & 状态转换 - LLM 分析代码生成测试用例

标准需求

REQ-001: 用户注册

测试技术

PathCoverage

StateTransition

生成测试用例

共 12 用例

路径覆盖 状态转换

覆盖率

关联行

107

覆盖行

107

覆盖率

100.0%

覆盖映射

TC-PC-001 用户注册成功

TC-PC-002 用户名已存在

ID	从	到	触发	守卫	动作
T1	S1	S2	提交注册请求	请求体为JSON格式	解析请求数据
T2	S2	S3	验证用户名	用户名有效	继续验证密码
T3	S2	S8	验证用户名	用户名无效	返回错误信息
T4	S3	S4	验证密码	密码有效	继续验证邮箱
T5	S3	S8	验证密码	密码无效	返回错误信息
T6	S4	S5	验证邮箱	邮箱有效	继续验证年龄
T7	S4	S8	验证邮箱	邮箱无效	返回错误信息
T8	S5	S6	验证年龄	年龄有效	检查唯一性
T9	S5	S8	验证年龄	年龄无效	返回错误信息
T10	S6	S7	检查唯一性	用户名和邮箱唯一	创建用户

All-States 路径 (6)

- S1 → S2 → S3 → S4 → S5 → S6 → S7 | 完整的成功注册路径
- S1 → S2 → S8 | 用户名验证失败路径
- S1 → S2 → S3 → S8 | 密码验证失败路径
- S1 → S2 → S3 → S4 → S8 | 邮箱验证失败路径
- S1 → S2 → S3 → S4 → S5 → S8 | 年龄验证失败路径
- S1 → S2 → S3 → S4 → S5 → S6 → S8 | 唯一性检查失败路径

Select test requirements

Select test technique: PathCoverage
State-Transition

The system generates: covered paths,
coverage results, and the state
transition table

Manual review and modification

Workflow

- 1. Requirement Input
- 2. Risk Analysis
- 3. Black-box Test Design
- 4. White-box Test Design
- 5. Oracle Generation
- 6. Suite Optimization
- 7. Export

预言生成

填写测试数据 → LLM 推导预期结果 → 保存到自定义套件

标准需求

REQ-005: 任务管理

测试数据

数据 (JSON)

```
{
  "title": "",
  "description": "",
  "priority": "",
  "status": ""
}
```

生成预期结果

预期结果

状态码

400

```
{
  "error": "任务标题不能为空",
  "field": "title"
}
```

根据需求中“任务CRUD:title必填<=200”的规则，任务标题是必填字段。如果标题为空，系统应返回400状态码，并提示用户“任务标题不能为空”的错误信息。代码中 validate_task_title 函数会验证标题是否为空，并返回相应的错误消息。

用例名称

TC-CUSTOM-REQ-005-002

保存到自定义套件

自定义套件

> 自定义套件 - 任务管理 (1)

The system generates expected results from the input test data

Generated test cases can be saved to a custom suite

Workflow

1. Requirement Input
2. Risk Analysis
3. Black-box Test Design
4. White-box Test Design
5. Oracle Generation
- 6. Suite Optimization**
7. Export

套件优化

有效/无效区管理·风险优先级·合并·覆盖最小化

测试套件

白盒测试套件-用户注册

全部	有效	无效
12	12	0

有效区 (12) 无效区 (0)

▼ [V] TC-PC-001 用户注册成功 [PathCoverage] (Valid)

测试所有输入字段有效时的注册流程

Tags: path_coverage

Step 1

//validate_username正常返回 → validate_password正常返回 → validate_email正常返回 → validate_age正常返回 → /register处理 → 返回201

```
{
  "username": "ValidUser123"
  "password": "Valid@Pass123!"
  "confirm_password": "Valid@Pass123!"
  "email": "valid_email@example.com"
}
```

自动优化

风险优先级·合并相同用例·覆盖最小化

保留比例

0.10 1.00 1.00

最低风险等级

全部

执行

全部恢复有效

清除状态记录

Test items can be manually enabled, disabled, and modified

The system can sort and filter test cases by risk priority, coverage relationships, and duplication

Workflow

1. Requirement Input
2. Risk Analysis
3. Black-box Test Design
4. White-box Test Design
5. Oracle Generation
6. Suite Optimization
- 7. Export**

导出

Pytest 可运行脚本 / JSON 结构化文档

套件	有效用例	全部用例
3	31	31

选择套件

白盒测试套件 - ... × 黑盒测试套件 - ... × 自定义套件 - ... ×

格式

☒ Pytest (可运行) ☐ JSON (文档)

导出

下载

导出完成

```
"""AutoTestDesign 自动生成 | 2026-06-03T10:33:05.044599"""
import pytest, json, sys, os

_root = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
sys.path.insert(0, _root)

from flask_app.app import create_app
```

格式

☐ Pytest (可运行) ☒ JSON (文档)

导出

下载

导出完成

```
{
  "test_suites": [
    {
      "id": "TS-WB-REQ-001",
      "name": "白盒测试套件 - 用户注册",
      "requirement_id": "REQ-001",
      "test_cases": [
        {
          "id": "TC-PC-001",
          "requirement_id": "REQ-001",
          "title": "用户注册成功",
          "description": "测试所有输入字段有效时的注册流程",
          "technique": "PathCoverage",
          "category": "Valid",
          "preconditions": [],

```

Test cases in the suite can be exported as JSON or as runnable Pytest

Video Demo

Risk Analysis

需求	≡ 风险分数	≡ 风险	≡ 优先级
REQ-001	88	High	High
REQ-002	92	High	High
REQ-003	85	High	High
REQ-004	70	Medium	Medium
REQ-005	65	Medium	Medium
REQ-006	75	Medium	Medium
REQ-007	50	Low	Low

REQ-001~003

High — authentication & account security

REQ-004~006

Medium — core project & task business

REQ-007

Low — read-only statistics logic

Test Design

Test Strategy

Adopt a risk-driven strategy:

High-risk requirements prioritize authentication security;

Medium-risk requirements focus on business rules and key paths;

Low-risk requirements focus on statistics logic and interface consistency.

Technique Combination

Black-box: EP / BVA / DT

White-box: path coverage / state transition

Test Plan

Select REQ-005, REQ-006, REQ-007

Compare three script types: auto-generated, manually verified, and hand-written

Experiment Results

Dimension	test_generated.py	test_generated_refined.py	test_human.py
Source	Auto-generated	Auto-generated, then manually verified	Hand-written from scratch
Test functions	44	32	27
Script's own errors	17	0	0
Real defects found	0	3 (BUG-7, BUG-8, BUG-9)	1 (BUG-9)
Fix / writing time	~3-5 min, auto-generated	~20 min verifying & fixing	~4 hrs of manual design & writing
LLM cost	Input 79,260 / Output 10,828, cost ≈ ¥2	Input 79,260 / Output 10,828, cost ≈ ¥2	¥0

Conclusion:

The main value of the automated tool is more thorough test perspectives, but it produces some of its own bugs.

Hand-written scripts are inherently reliable, but their test perspectives are not rich enough.

Comparison Experiment

Defect	Meaning	Found this round
BUG-7	Assignment/update for an unregistered owner not handled	Found by <code>test_generated_refined.py</code>
BUG-8	Completed tasks can still be deleted	Found by <code>test_generated_refined.py</code>
BUG-9	A past <code>due_date</code> is not rejected	Found by <code>test_generated_refined.py</code> and <code>test_human.py</code>

As can be seen, automated scripts after manual verification clearly outperform hand-written scripts

Anomaly 1: REQ-005 coverage is only 43.8%

覆盖率 GO

关联行	覆盖行	覆盖率
73	32	43.8%

2 代码段未覆盖

routes/tasks.py L25-37 列出项目下的所有任务

routes/tasks.py L61-68 获取单个任务详情

覆盖率 GO

关联行	覆盖行	覆盖率
73	73	100%

Original result:

associated lines 73, covered lines 32, coverage 43.8%

Investigation found:

The requirement side focused on create, update, and delete, while downplaying the list and detail paths

The requirement-code map pulled in a broad range of task-routing code

Coverage is counted by line, while the uncovered-line hints only show fully uncovered segments

After updating the requirement description:

covered lines rose to 73 lines, coverage became 100%

Anomaly 2: In black-box testing, EP/BVA did not generate test cases

REQ-007 definition:

input_fields is an empty array;

conditions is an empty array;

expected_behaviors has only one: "exists -> returns statistics result"

EP: No explicit set of input fields, and no field-level valid / invalid classes

BVA: No numeric range, length range, or explicit boundary

**Not suitable for
case generation**

The current tool shows some technical adaptability to different requirement types:

when a requirement lacks field-boundary information, the tool does not force out low-quality EP/BVA cases, but is more likely to produce results on decision tables and state transitions

Performance & Generalization

Model	Configuration	Total time	Valid cases	EP	BVA	DT	Verdict
gpt-4o	yunwu.ai / reasoning param disabled	24.414s	22	7.615s / 12 cases	11.060s / 6 cases	5.739s / 4 cases	Most stable this round
deepseek-v4-flash	thinking disabled	108.415s	3	43.897s / 0 cases	60.405s / 0 cases	4.113s / 3 cases	DT works, EP/BVA unstable
gemini-3.5-flash	free API	70.940s	14	16.245s / 8 cases	18.400s / 6 cases	36.295s / 0 cases	EP/BVA works, DT unstable

In terms of generation speed and completeness, under the current configuration, choosing gpt-4o is more appropriate.

Performance & Generalization

代码仓库

路径

flask_app

13 个 .py 文件

需求文档

导入方式

☒ CSV 标准需求 ☐ TXT 文本需求 ☐ 粘贴文本

上传预定义的标准需求 CSV

Upload

200MB per file • CSV

This system supports custom projects and generates test cases directly from requirements and the project repository

Project Summary

AutoTestDesign establishes a traceable design chain from requirements to test cases, code paths, and execution scripts

01

Clear advantages in the test-design stage

It can quickly produce black-box analysis tables, white-box coverage ideas, and large-scale candidate test scripts

02

Auto-generate a first draft, with humans doing the key verification

This approach strikes a better balance between time cost and defect-detection capability.

03

Auto-generated scripts cannot completely skip the manual step

The raw auto-exported scripts had many problems, but after one modest round of manual verification, script quality improved markedly and defect-detection ability was significantly enhanced.

小组分工

2351269黄一和

Responsible for building the overall code structure of the project; implemented requirement input, CSV / text parsing, and standardized requirement modeling; organized the mapping between requirements and code

2352744关镜文

Responsible for risk analysis, black-box and white-box test design; designed and tuned LLM Prompt; implemented EP, BVA, decision tables, path coverage, state transition, and other generation flows

2351271黄弋涵

Responsible for the target Flask application and test-script execution; maintained the Pytest export and run workflow; ran comparison experiments across hand-written, automated, and verified scripts; analyzed defect detection and anomalies

2352048夏浩博

Responsible for Streamlit interface optimization and workflow presentation; organized screenshot materials; unified document format and wording; created the PPT



感谢观看
恳请老师批评指正